C A P S T O N E P R O J E C T R E P O R T

SQL for Data Science

Capstone Project

CoffeeKing: Data-Driven Nashville Coffee Market Analysis

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Platform	Coursera
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Certificate ID	9G82IHYVPIL8
Completion Date	June 3, 2026
Client	CoffeeKing (fictitious Nashville coffee startup)
Dataset	Yelp Academic Dataset, Filtered for Nashville, TN
Tool	Databricks Free Edition, SQL Notebook (18 SQL cells)
Milestones	4 completed: Proposal, Descriptive Stats, Deeper Analysis, Storytelling

Executive Summary

This report documents the complete end-to-end data analysis project completed as part of the SQL for Data Science Capstone Project course offered by the University of California, Davis through Coursera. The project was delivered across four structured milestones over the course of the programme, progressing from initial data preparation and proposal through descriptive statistics, correlation analysis, new metric creation, and finally a full storytelling presentation to a fictional client management team.

The client selected for this project is CoffeeKing, a fictitious startup coffee company seeking to establish its first physical presence in Nashville, Tennessee. The analytical objective was to provide CoffeeKing with data-driven recommendations on optimal location selection, operating hours strategy, quality benchmarks, and competitive positioning, using the Yelp Academic Dataset filtered for Nashville as the primary data source.

Across 18 SQL queries executed in a Databricks Free Edition notebook, the analysis covered 394 Nashville coffee and cafe businesses, 919,271 individual Yelp reviews, 54,115 tips, 6,178 check-in records, and 180,431 user profiles. The analysis produced three major findings: downtown zip code 37203 is oversaturated with low quality perception, the Mon-Sat operating schedule significantly outperforms 7-day operations on customer rating, and zip code 37216 in East Nashville consistently ranks as the optimal target location across every independent analysis method applied.

Top recommendation: CoffeeKing should open its first location in zip code 37216 (East Nashville), operate Monday to Saturday, target a minimum 4.0 star rating from day one, and benchmark against Crema Coffee Roasters (Quality Score 13.50) as the market leader to emulate and eventually surpass.

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Section 1
Project Background and Setup

1. Project Background and Setup

1.1 Course Overview

The SQL for Data Science Capstone Project is a graduate-level applied analytics course offered by the University of California, Davis through the Coursera platform. The course is designed to develop practical SQL-based data science skills through a real-world simulation in which the student assumes the role of a data scientist at a data analytics firm, tasked with delivering strategic business insights to a client of their choosing.

The course is structured around four sequential milestones that mirror a real professional data analytics engagement: scoping and proposal, exploratory analysis and descriptive statistics, deeper analytical techniques and new metric creation, and finally the storytelling and presentation of findings to management. Each milestone builds directly on the previous one, creating a coherent and progressively sophisticated analytical narrative.

1.2 Client Selection: CoffeeKing

Three fictitious client companies were available for selection: CoffeeKing (Yelp Nashville restaurant data), Lobbyists4America (Congressional tweets 2008 to 2017), and SportsStats (120 years of Olympic data). CoffeeKing was selected for the following three reasons:

- Richness and accessibility of the data: The Yelp Academic Dataset provides five interconnected relational tables covering business attributes, customer reviews, tips, check-in records, and user profiles. These tables contain a diverse range of data types including numeric ratings, text reviews, JSON-formatted attributes and hours, and timestamp data, making them ideal for structured SQL analysis across multiple analytical techniques.
- Clarity of business questions: The client brief for CoffeeKing centred on location selection, hours of operation, and competitive positioning. These are concrete, measurable questions that can be answered directly through SQL aggregation, filtering, and JOIN operations without requiring complex natural language processing or domain-specific knowledge.
- Professional relevance: The hospitality and consumer review sector is a well-understood domain that produces findings easily communicable to a non-technical management audience, directly satisfying the storytelling requirements of Milestone 4.

1.3 Analytical Tool: Databricks Free Edition

All SQL analysis was conducted using Databricks Free Edition, a cloud-based notebook environment that supports SQL, Python, R, and Scala within a single unified workspace.

Databricks was selected over alternatives such as DB Browser for SQLite, DBeaver, or VS Code because it provides a notebook interface in which each SQL cell retains both the query code and its results simultaneously, enabling clean screenshots for documentation, persistent session storage across queries, built-in result visualisation capabilities, and industry-standard infrastructure used by major technology companies including Netflix, Apple, Uber, and thousands of others globally. The five Yelp CSV files were uploaded directly to Databricks and converted to Delta tables, after which all 18 SQL queries were executed within a single notebook titled CoffeeKing EDA Cleaning.

Section 2
Dataset Description and Entity Relationship Diagram

2. Dataset Description and Entity Relationship Diagram

2.1 Dataset Overview

The Yelp Academic Dataset (Nashville, TN filter applied) was provided as a ZIP archive containing five CSV files. After import into Databricks, the dataset resolved into the following five relational tables:

Table name	Row count	Primary key	Foreign keys	Key columns
business	6,971	business_id	None	name, stars, review_count, categories, hours, postal_code, is_open, attributes
review	919,271	review_id	business_id, user_id	stars, text, date, useful, funny, cool
tip	54,115	None	business_id, user_id	text, date, compliment_count
checkin	6,178	None	business_id	date (comma-separated multi-value list)
user	180,431	user_id	None	name, average_stars, review_count, fans, yelping_since, elite

2.2 Entity Relationship Diagram

The five tables form a star-schema structure centred on the BUSINESS table. All four satellite tables connect to BUSINESS via business_id as a foreign key. The USER table additionally connects to both REVIEW and TIP via user_id, enabling analysis of reviewer behaviour across both review types. The CHECKIN table links solely to BUSINESS and contains multi-value date strings rather than one row per visit.

ERD relationship summary: BUSINESS (1) to REVIEW (many) via business_id. BUSINESS (1) to TIP (many) via business_id. BUSINESS (1) to CHECKIN (1 row) via business_id. USER (1) to REVIEW (many) via user_id. USER (1) to TIP (many) via user_id.

2.3 Data Quality Notes

Several structural data quality characteristics were identified during the profiling phase that required acknowledgement and handling throughout the analysis:

- _rescued_data column: All five tables contained a column named _rescued_data, automatically generated by the Databricks ingestion pipeline as a safety net for rows containing values that did not conform to the inferred schema. In this dataset all values in the column were empty strings or NULL, confirming that the original CSV files were clean at the schema level. The column was excluded from all queries.
- postal_code data type: The postal_code column was inferred as bigint (integer) by Databricks rather than string. This is technically incorrect for postal codes because leading zeros in codes such as 07030 would be silently dropped. CAST(postal_code AS STRING) was applied in all location-based queries to preserve correct formatting.
- hours and attributes columns: Both columns contain JSON-formatted strings embedded within CSV cells rather than normalised relational data. For example, hours stores values such as {Monday: 8:0-17:0, Tuesday: 8:0-17:0}. Full JSON parsing was not performed within the scope of this project. Instead, SQL LIKE pattern matching was used to extract relevant patterns such as opening times and specific attribute values.
- address nulls: 285 of 6,971 business records (4.1%) had no address value. These records were excluded from any address-dependent queries but retained for all other analyses.
- Chain business contamination: A small number of chain fast food and retail businesses including McDonald's, Starbucks, and Krispy Kreme appeared in the coffee and cafe category filter. These were identified through the cross-table JOIN analysis in Milestone 3 and noted for exclusion in future refined analyses.

Section 3
Data Cleaning and Quality Assessment

3. Data Cleaning and Quality Assessment

Data cleaning was conducted systematically before any analytical queries were executed. The following seven steps were applied in sequence, each documented with a corresponding SQL cell in the Databricks notebook:

Step 1: Row Count Profiling

The first SQL query counted all rows in each of the five tables using UNION ALL to produce a single consolidated result. This confirmed successful data loading and established the baseline record counts that appear throughout this report.

```
SELECT 'business' AS table_name, COUNT(*) AS row_count FROM business
UNION ALL
SELECT 'review',  COUNT(*) FROM review
UNION ALL
SELECT 'tip',     COUNT(*) FROM tip
UNION ALL
SELECT 'checkin', COUNT(*) FROM checkin
UNION ALL
SELECT 'user',    COUNT(*) FROM user;
```

Step 2: Column Profiling with DESCRIBE TABLE

The DESCRIBE TABLE command was executed on the business table to retrieve all column names and their inferred data types. This confirmed 15 columns, identified the postal_code bigint issue, noted the presence of the _rescued_data artifact column, and confirmed that hours and attributes were stored as string type rather than a native JSON type, informing the LIKE-based extraction approach used in subsequent queries.

```
-- STEP 2: Explore columns in business table
DESCRIBE TABLE business;
```

Step 3: NULL Value Assessment

A comprehensive NULL check was executed using COUNT() on each critical column. COUNT() in SQL counts only non-NULL values, so any column returning fewer than the total row count indicates missing data. Results confirmed that address had 285 NULLs (6,686 non-null out of 6,971), postal_code had 4 NULLs (6,967 non-null), and all other key columns (stars, categories, hours, is_open, review_count, business_id, name) were fully populated.

```
-- STEP 3: Check NULLs in business table
```

```
SELECT
  COUNT(*) AS total_rows,
  COUNT(business_id) AS has_business_id,
  COUNT(name) AS has_name,
  COUNT(address) AS has_address,
  COUNT(stars) AS has_stars,
  COUNT(categories) AS has_categories,
  COUNT(hours) AS has_hours,
  COUNT(postal_code) AS has_postal_code,
  COUNT(is_open) AS has_is_open
FROM business;
```

Step 4: Star Rating Validation

A GROUP BY query on the stars column of the business table was used to enumerate all distinct values and their frequencies. This validated that the data contained only the nine expected Yelp star values (1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, and 5.0) with no outliers, rogue values, or data entry errors. The distribution showed 4.0 stars as the most common rating with 1,493 businesses, and 1.0 star as the rarest with only 99 businesses.

```
-- STEP 4: Validate star ratings in business
SELECT
  stars,
  COUNT(*) AS count
FROM business
GROUP BY stars
ORDER BY stars;
```

Step 5: Coffee Category Filter

A WHERE clause using four LOWER(categories) LIKE conditions was developed to isolate coffee and cafe businesses from the full 6,971-row dataset. The LOWER() function was applied to ensure case-insensitive matching, capturing variations such as Coffee, coffee, Cafe, cafe, Cafes, and Coffee Roasters. This filter consistently returned 394 businesses across all subsequent queries and was used as the standard coffee subset for all analysis in Milestones 1 through 4.

```
-- STEP 5: Count coffee/cafe businesses
SELECT COUNT(*) AS total_coffee_businesses
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
   OR LOWER(categories) LIKE '%cafes%'
   OR LOWER(categories) LIKE '%coffee roasters%';
```

Step 6: Coffee Subset Summary Statistics

A single aggregated query computed key summary statistics for the 394-business coffee subset: total count, average star rating (3.7), minimum and maximum stars (1.0 and 5.0), total reviews across all shops (37,881), and a breakdown of currently open (266) versus permanently closed

(128) businesses. The 33% closure rate was an immediate and significant finding that became a central theme in the competitive analysis.

```
-- STEP 6: Count coffee/cafe businesses summary
SELECT
  COUNT(*) AS total_coffee_businesses,
  ROUND(AVG(stars), 2) AS avg_stars,
  MIN(stars) AS min_stars,
  MAX(stars) AS max_stars,
  SUM(review_count) AS total_reviews,
  SUM(CASE WHEN is_open = 1 THEN 1 ELSE 0 END) AS currently_open,
  SUM(CASE WHEN is_open = 0 THEN 1 ELSE 0 END) AS permanently_closed
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
   OR LOWER(categories) LIKE '%cafes%'
   OR LOWER(categories) LIKE '%coffee roasters%';
```

Step 7: Data Type Correction

CAST(postal_code AS STRING) was confirmed as the standard approach for all location-based queries. This was validated by observing that the raw bigint representation produced correct 5-digit outputs for all Nashville zip codes in the dataset, with the CAST ensuring string-based GROUP BY operations produced correctly formatted zip code labels in all output tables.

```
-- STEP 7: Validate postal_code CAST to STRING
SELECT
  CAST(postal_code AS STRING) AS zip_code,
  COUNT(*) AS shop_count
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
GROUP BY postal_code
ORDER BY shop_count DESC
LIMIT 5;
```

Section 4
Milestone 1: Project Proposal

4. Milestone 1: Project Proposal

4.1 Project Description

CoffeeKing is a new startup coffee company aiming to provide a unique and novel customer experience in Nashville, Tennessee, and wants to appeal to a wide variety of clientele. The company requires data-driven guidance on location selection, hours of operation, and competitive positioning before committing to a physical site. This project uses the Yelp Academic Dataset to answer three core research questions that address these strategic priorities directly.

4.2 Research Questions

1. **Q1.** Which Nashville zip codes have the highest average star ratings for coffee shops, and how does review volume and closure rate vary by location? This question directly informs CoffeeKing's location selection strategy by identifying which markets are both high-quality and commercially stable.
2. **Q2.** Is there a relationship between a coffee shop's review volume and its average star rating? Do businesses with more reviews tend to score higher or regress toward a mid-range mean?
3. **Q3.** What operating schedule do top-rated Nashville coffee shops share? Do businesses open more days per week earn higher or lower customer ratings than those with focused, restricted schedules?

4.3 Initial Hypotheses

Three testable hypotheses were formulated before any analysis was conducted, based on general market intuition and prior knowledge of urban consumer behaviour:

4. **H1.** Downtown Nashville zip codes such as 37201 and 37203 likely have more coffee businesses due to tourist and commuter foot traffic, but increased competition and high customer volume would result in lower average ratings compared to more residential or neighbourhood zip codes.
5. **H2.** Coffee shops with very high review counts (500 or more reviews) will tend to regress toward the mean, clustering around 3.5 to 4.0 stars, while lesser-known shops with fewer reviews may show more extreme ratings at either end of the scale.
6. **H3.** Coffee shops open all seven days per week will earn higher customer ratings than those with restricted schedules, as greater availability and consistency of access correlates positively with customer satisfaction and convenience.

4.4 Key Findings from Milestone 1

The following results were produced from SQL Cells 6 through 11 in the Databricks notebook:

Location Analysis (Q1 Answer)

```
-- CELL 8: Q1 - Best zip codes for coffee shops
SELECT
  CAST(postal_code AS STRING) AS zip_code,
  COUNT(*) AS total_shops,
  ROUND(AVG(stars), 2) AS avg_stars,
  SUM(review_count) AS total_reviews,
  SUM(CASE WHEN is_open = 1 THEN 1 ELSE 0 END) AS open_shops,
  SUM(CASE WHEN is_open = 0 THEN 1 ELSE 0 END) AS closed_shops
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
GROUP BY postal_code
HAVING COUNT(*) >= 3
ORDER BY avg_stars DESC, total_reviews DESC;
```

Zip code	Total shops	Avg stars	Total reviews	Open shops	Closed shops
37220	3	4.67	93	2	1
37206	**31**	**4.24**	**4,125**	**19**	**12**
37216	**13**	**4.19**	**626**	**9**	**4**
37208	19	4.16	3,102	10	9
37203	78	3.65	7,034	56	22

H1 SUPPORTED: Zip 37203 has the most shops (78) but one of the lowest average ratings (3.65). High competition and tourist volume create a race to the bottom on quality perception. Zip 37206 emerges as the first strong candidate with 4.24 stars and 31 shops.

Review Volume vs Rating (Q2 Answer)

```
-- CELL 9: Q2 - Review volume vs star rating
SELECT
  CASE
    WHEN review_count < 50 THEN '1. Low (under 50)'
    WHEN review_count < 200 THEN '2. Mid (50-199)'
    WHEN review_count < 500 THEN '3. High (200-499)'
    ELSE '4. Very High (500+)'
  END AS review_bucket,
  COUNT(*) AS shop_count,
  ROUND(AVG(stars), 2) AS avg_stars,
  ROUND(PERCENTILE(review_count, 0.5), 0) AS median_reviews
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
```

```
GROUP BY review_bucket
ORDER BY review_bucket;
```

Review bucket	Shop count	Avg stars	Median reviews
Low (under 50 reviews)	231	3.53	23
Mid (50 to 199 reviews)	116	3.89	98
High (200 to 499 reviews)	34	4.04	299
Very High (500+ reviews)	**13**	**4.12**	**859**

H2 DISPROVED: Opposite of the hypothesis is true. Shops with 500 or more reviews score the highest average (4.12 stars). Quality and popularity rise together rather than regressing to the mean. This became one of the most important strategic findings of the entire project.

Operating Schedule vs Rating (Q3 Answer)

```
-- CELL 10: Q3 - Opening hours vs star rating
SELECT
  CASE
    WHEN hours LIKE '%"Monday":"%'
     AND hours LIKE '%"Saturday":"%'
     AND hours LIKE '%"Sunday":"%' THEN '7 days a week'
    WHEN hours LIKE '%"Monday":"%'
     AND hours LIKE '%"Saturday":"%' THEN 'Mon-Sat (6 days)'
    WHEN hours LIKE '%"Monday":"%' THEN 'Weekdays only'
    ELSE 'Other/Irregular'
  END AS schedule_type,
  COUNT(*) AS shop_count,
  ROUND(AVG(stars), 2) AS avg_stars,
  SUM(CASE WHEN is_open = 1 THEN 1 ELSE 0 END) AS currently_open
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
GROUP BY schedule_type
ORDER BY avg_stars DESC;
```

Schedule type	Shop count	Avg stars	Currently open
Mon-Sat (6 days)	**37**	**4.26**	**21**
Weekdays only	22	4.05	10
Other/Irregular	56	3.84	24
7 days a week	279	3.57	211

H3 DISPROVED: Mon-Sat shops score 4.26 average stars versus 3.57 for 7-day shops. Focused independent specialty cafes dramatically outperform high-volume chain-style 7-day operations on customer satisfaction. Strategic recommendation: open 6 focused days, not 7 mediocre ones.

Section 5
Milestone 2: Descriptive Statistics

5. Milestone 2: Descriptive Statistics and Understanding the Data

5.1 Five-Number Summary

Milestone 2 applied formal descriptive statistical analysis to the 394 Nashville coffee businesses using SQL aggregate functions including AVG, STDDEV, MIN, MAX, and PERCENTILE. The five-number summary (minimum, Q1, median, Q3, maximum) was computed for both the star rating and review count variables.

```
-- CELL 12: Full descriptive statistics
SELECT
  COUNT(*) AS total_shops,
  ROUND(AVG(stars), 3) AS mean_stars,
  ROUND(STDDEV(stars), 3) AS stddev_stars,
  MIN(stars) AS min_stars,
  MAX(stars) AS max_stars,
  ROUND(PERCENTILE(stars, 0.25), 1) AS q1_stars,
  ROUND(PERCENTILE(stars, 0.50), 1) AS median_stars,
  ROUND(PERCENTILE(stars, 0.75), 1) AS q3_stars,
  ROUND(AVG(review_count), 1) AS mean_reviews,
  ROUND(STDDEV(review_count), 1) AS stddev_reviews,
  MIN(review_count) AS min_reviews,
  MAX(review_count) AS max_reviews,
  ROUND(PERCENTILE(review_count, 0.50), 0) AS median_reviews
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%';
```

Statistic	Star ratings	Review counts per shop
Mean (average)	3.702	96.1
Standard deviation	0.923	174.4
Minimum	1.0	5
Q1 (25th percentile)	3.0	approx. 18
Median (50th perc.)	4.0	39
Q3 (75th percentile)	4.5	approx. 98
Maximum	5.0	1,725

5.2 Analytical Interpretation

The mean star rating of 3.702 and median of 4.0 are close, indicating mild left skew in the distribution. Most Nashville coffee shops cluster at 4.0 stars with a relatively small proportion

rated below 2.5. The standard deviation of 0.923 confirms that the market is genuinely stratified, with approximately two thirds of all shops falling between 2.78 and 4.62 stars.

The review count variable tells a fundamentally different story. The mean of 96.1 reviews is more than 2.5 times the median of 39, revealing strong right skew caused by a handful of landmark businesses with hundreds or thousands of reviews pulling the average upward. Milk and Honey Nashville alone holds 1,725 reviews while the typical Nashville coffee shop holds only 39. This pattern means that mean review counts are a misleading metric, and all review-based analysis in this project uses median as the primary measure of central tendency.

5.3 Star Tier Analysis

Cell 13 in the notebook grouped coffee businesses into four star rating tiers and compared their review volume statistics. This produced a counterintuitive and strategically important discovery:

```
-- CELL 13: Star tier analysis with review volume
SELECT
  CASE
    WHEN stars >= 4.5 THEN '4.5 to 5.0 (top tier)'
    WHEN stars >= 4.0 THEN '4.0 to 4.4 (good)'
    WHEN stars >= 3.0 THEN '3.0 to 3.9 (average)'
    ELSE 'below 3.0 (poor)'
  END AS star_tier,
  COUNT(*) AS shop_count,
  ROUND(AVG(review_count), 1) AS avg_reviews,
  ROUND(PERCENTILE(review_count, 0.5), 0) AS median_reviews,
  SUM(CASE WHEN is_open = 1 THEN 1 ELSE 0 END) AS open_shops
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
GROUP BY star_tier
ORDER BY avg_reviews DESC;
```

Star tier	Shops	Avg reviews	Median reviews	Currently open
4.0 to 4.4 (good)	**110**	**146.5**	**56**	**64**
4.5 to 5.0 (top tier)	126	99.3	42	86
3.0 to 3.9 (average)	92	78.7	33	61
Below 3.0 (poor)	66	30.5	30	55

Discovery: The 4.0 to 4.4 tier generates more average reviews (146.5) than the 4.5 to 5.0 top tier (99.3). This reveals that the most engaged, most visited businesses are not the highest rated. Popular busy shops receive more diverse ratings from a larger customer base, while top-tier shops tend to be smaller specialty venues with loyal but smaller audiences.

```
-- CELL 14: Zip code survival rate analysis
SELECT
```

```
  CAST(postal_code AS STRING) AS zip_code,
  COUNT(*) AS total_shops,
  ROUND(AVG(stars), 2) AS avg_stars,
  ROUND(PERCENTILE(stars, 0.5), 1) AS median_stars,
  SUM(CASE WHEN is_open = 1 THEN 1 ELSE 0 END) AS open_shops,
  ROUND(SUM(CASE WHEN is_open = 1 THEN 1 ELSE 0 END) * 100.0
        / COUNT(*), 1) AS survival_rate_pct
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
GROUP BY postal_code
HAVING COUNT(*) >= 5
ORDER BY survival_rate_pct DESC, avg_stars DESC;
```

A further discovery emerged from the survival rate analysis: zip codes with the highest business survival rates do not necessarily have the highest ratings. Zip codes 37207 and 37217 both showed 87.5% survival rates but average ratings of only 3.25 and 3.0 respectively, confirming that foot traffic and low competition can sustain a mediocre business, but quality is required for customer satisfaction. This underscored the importance of the composite Market Opportunity Index developed in Milestone 3.

Section 6
Milestone 3: Beyond Descriptive Stats

6. Milestone 3: Beyond Descriptive Stats (Dive Deeper / Go Broader)

6.1 Pearson Correlation Analysis

Cell 15 applied SQL's built-in CORR() function to compute Pearson correlation coefficients between three pairs of variables across the 394 coffee businesses. Pearson correlation measures the linear relationship between two continuous variables, returning a value between -1.0 (perfect negative relationship) and +1.0 (perfect positive relationship), with values near 0 indicating no linear relationship.

```
-- CELL 15: Pearson correlation
SELECT
  ROUND(CORR(stars, review_count), 4) AS corr_stars_reviews,
  ROUND(CORR(stars, is_open), 4) AS corr_stars_isopen,
  ROUND(CORR(review_count, is_open), 4) AS corr_reviews_isopen,
  COUNT(*) AS total_shops
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%';
```

Variable pair	Correlation coefficient	Interpretation	Strategic implication
Stars vs review_count	+0.1587	Weak positive	Quality attracts reviews; reviews do not create quality
Stars vs is_open	-0.1186	Weak negative (surprising)	High ratings alone do not guarantee business survival
Review count vs is_open	+0.1435	Weak positive	More visible businesses survive slightly better

Key analytical finding: All three Pearson correlations are weak (absolute values below 0.20). This means no single variable dominates the others. Coffee shop success in Nashville is determined by the interaction of multiple factors simultaneously, which justifies the composite metric approach developed in this milestone.

6.2 New Metric 1: Quality Score

Formula: Quality Score = stars x LOG10(review_count + 1)

Rationale: A business with 5.0 stars and only 2 reviews is not as proven or trustworthy as a business with 4.5 stars and 800 reviews. The Quality Score rewards both dimensions simultaneously. The LOG10 function compresses the review count scale, preventing businesses with thousands of reviews from completely dominating the ranking. Adding 1 before taking the logarithm ensures businesses with zero reviews produce a score of 0 rather than an undefined value.

```
-- CELL 16: New Metric 1 - Quality Score
SELECT
  name,
  stars,
  review_count,
  CAST(postal_code AS STRING) AS zip_code,
  is_open,
  ROUND(stars * LOG10(review_count + 1), 3) AS quality_score
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
ORDER BY quality_score DESC
LIMIT 15;
```

Top 5 Quality Score results:

Rank	Business name	Stars	Reviews	Zip	Quality score
1	Crema Coffee Roasters	4.5	998	37210	13.498
2	Milk and Honey Nashville	4.0	1,725	37203	12.948
3	The Garden Brunch Cafe	4.5	712	37208	12.839
4	Fido	4.0	1,260	37212	12.403
5	Sunflower Cafe	4.5	561	37204	12.374

6.3 New Metric 2: Market Opportunity Index

Formula: Opportunity Index = (avg_stars / 5.0) x 0.5 + (1.0 - shop_count / 100.0) x 0.3 + (survival_rate) x 0.2

Rationale: This composite index combines three strategic factors into a single ranked score per zip code. Quality of the existing market (weighted 50%) tells CoffeeKing whether customers in that area appreciate and reward good coffee. Competition level (weighted 30%) identifies zip codes with room for a new entrant to be noticed. Market stability (weighted 20%) confirms that the area can sustain a business over time. The weights reflect the strategic priority order for a new market entrant: quality market first, space to enter second, then stability.

```
-- CELL 17: New Metric 2 - Market Opportunity Index
SELECT
  CAST(postal_code AS STRING) AS zip_code,
  COUNT(*) AS total_shops,
  ROUND(AVG(stars), 2) AS avg_stars,
  ROUND(SUM(CASE WHEN is_open=1 THEN 1.0 ELSE 0 END)
        / COUNT(*) * 100, 1) AS survival_pct,
  ROUND(
    (AVG(stars) / 5.0) * 0.5 +
    (1.0 - COUNT(*) / 100.0) * 0.3 +
    (SUM(CASE WHEN is_open=1 THEN 1.0 ELSE 0 END)
     / COUNT(*)) * 0.2
  , 4) AS opportunity_index
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
GROUP BY postal_code
HAVING COUNT(*) >= 3
ORDER BY opportunity_index DESC
LIMIT 10;
```

Rank	Zip code	Avg stars	Shops	Survival rate	Opportunity index
1	37220	4.67	3	66.7%	0.8910
2	**37216**	**4.19**	**13**	**69.2%**	**0.8187**
3	37210	3.68	11	72.7%	0.7806
4	37215	3.63	15	80.0%	0.7783
9	37206	4.24	31	61.3%	0.7538
10	37203	3.65	78	71.8%	0.7510

Zip 37216 ranks 2nd on the Opportunity Index (0.8187) and consistently outperforms across all four milestones. Zip 37220 ranks first but has only 3 total shops, which is too small a market to validate the index reliably. Zip 37206 drops to 9th due to higher competition (31 shops) despite strong 4.24 avg stars.

6.4 Cross-Table JOIN Analysis

Cell 18 executed the first JOIN query of the project, connecting the business table to the review table via business_id. This cross-table analysis computed the star_gap metric: the arithmetic difference between Yelp's published business-level star rating and the mean of all individual review-level star ratings for that business.

```
SELECT b.name, b.stars AS business_stars,
       ROUND(AVG(r.stars), 2) AS avg_review_stars,
       ROUND(b.stars - AVG(r.stars), 2) AS star_gap,
       COUNT(r.review_id) AS review_count
FROM business b
JOIN review r ON b.business_id = r.business_id
```

```
WHERE LOWER(b.categories) LIKE '%coffee%'
GROUP BY b.name, b.stars, b.postal_code
HAVING COUNT(r.review_id) >= 50
ORDER BY ABS(b.stars - AVG(r.stars)) DESC
LIMIT 15;
```

The maximum star gap found was only 0.29, confirming that Yelp's published business star ratings are accurate within a narrow margin of the true individual review average. This validation is important for the integrity of the entire analysis, as all location and quality recommendations are built on the business-level star ratings. The analysis also identified chain businesses (McDonald's, Starbucks, Krispy Kreme) appearing in the coffee category filter, a data quality issue flagged for exclusion in future refined analyses.

Section 7
Milestone 4: Presenting Findings (Storytelling)

7. Milestone 4: Presenting Findings (Storytelling)

7.1 Presentation Approach

The final milestone required translating the technical SQL analysis into a compelling, audience-appropriate presentation for CoffeeKing's senior management team and potential investors. The audience was identified as senior management and investors who need strategic, actionable findings to support a location and funding decision, rather than technical reviewers interested in SQL methodology.

The presentation was designed as a 16-slide PowerPoint deck built programmatically using the PptxGenJS library in Node.js. This approach ensured consistent professional formatting, a coherent colour scheme (navy, blue, teal, and green), and reproducible output. The deck includes a title slide, agenda, section dividers, metric cards, native PowerPoint charts, data tables, and a closing recommendations slide.

7.2 Presentation Structure

The 16-slide deck was structured around six narrative sections, following the principle of starting at the 10,000-foot level and working progressively toward the specific, actionable recommendations:

- Section 01: Project Overview. Client profile, dataset summary, and the three initial hypotheses before they were tested.
- Section 02: Data Exploration. The five-table schema, data cleaning steps, and the ERD illustrating table relationships.
- Section 03: Descriptive Statistics. Five-number summary and the star rating distribution bar chart showing 53% of shops at 4.0 or above.
- Section 04: Deeper Analysis. Pearson correlation results, Quality Score rankings, and Opportunity Index zip code rankings.
- Section 05: Hypotheses Right and Wrong. A colour-coded table showing H1 supported and H2/H3 disproved, with evidence and next steps.
- Section 06: Recommendations. A dedicated location recommendation slide for zip 37216, an operations strategy slide with bar chart, a competitive landscape table, and a final five-point recommendation summary on a dark background for visual impact.

7.3 Key Design Decisions

Several specific design choices were made to serve the management audience effectively. Charts were used in preference to tables wherever data shape mattered more than specific values. The star rating distribution was rendered as a native PowerPoint bar chart with colour-coded bars to distinguish poor, average, and good rating bands visually. The Market Opportunity Index results were displayed as a ranked table with green highlighting on the top recommendation to make the primary call to action immediately obvious. The hypothesis outcomes slide used traffic-light colour coding (green for supported, orange for disproved) to communicate results at a glance without requiring the audience to read detailed text.

Section 8
Final Recommendations for CoffeeKing

8. Final Recommendations for CoffeeKing

Based on the complete analysis across all four milestones and 18 SQL queries, the following five evidence-based recommendations are made for CoffeeKing's Nashville market entry:

7. **Rec 1.** Open in zip code 37216 (East Nashville). This recommendation is confirmed by four independent analysis methods: second-highest average star rating among zip codes with meaningful competition (4.19 stars), 69.2% business survival rate, only 13 existing competitors, and second-highest Market Opportunity Index score (0.8187). Zip 37216 consistently outperforms alternatives in every analytical framework applied across three milestones.

8. **Rec 2.** Operate Monday to Saturday. The data clearly shows that Mon-Sat shops score 4.26 average stars versus 3.57 for 7-day shops, a gap of 0.69 stars. Focused independent specialty cafes with shorter operating windows deliver dramatically better customer experiences than high-volume 7-day chain-style operations. Sunday closure should be used for staff development, quality review, and supplier relationships.

9. **Rec 3.** Set a minimum target of 4.0 stars from opening day. The market median is 4.0 stars, and 53.3% of all Nashville coffee shops already achieve this level. Any rating below 4.0 places CoffeeKing in the bottom half of the market. Investing in staff training, ingredient quality, and consistent preparation standards before opening is more important than early expansion or marketing.

10. **Rec 4.** Avoid downtown zip code 37203. With 78 competing coffee businesses and an average rating of only 3.65 stars, downtown Nashville is simultaneously the most competitive and the lowest-quality perception zone in the city. High tourist foot traffic does not compensate for the intense price and quality pressure in this area. CoffeeKing would be entering a race to the bottom rather than a market it can differentiate within.

11. **Rec 5.** Study and benchmark against Crema Coffee Roasters (zip 37210). Crema holds the highest Quality Score in the dataset at 13.498, combining 4.5 stars with 998 reviews. It represents the gold standard for Nashville independent coffee: high quality, high engagement, proven over time. CoffeeKing should set a target Quality Score of 12.0 or above within 18 months of opening, equivalent to approximately 4.3 stars with 300 or more reviews.

Summary: Open in 37216. Open Mon to Sat. Target 4.0+ stars from day one. Avoid downtown. Beat Crema Coffee Roasters. These five actions, if executed consistently, give CoffeeKing the highest probability of long-term success based on the evidence in the Nashville Yelp dataset.

Section 9
Hypothesis Review Summary

9. Hypothesis Review Summary

All three hypotheses proposed in Milestone 1 were rigorously tested using SQL aggregate queries in Milestones 1 through 3. The outcomes are summarised below. It is important to note that two disproved hypotheses represent a strong analytical result, not a failure. The data revealed real, non-obvious patterns that redirect strategic thinking toward truth.

Hypothesis	Verdict	Evidence	Implication
H1: Downtown 37203 has more shops but lower ratings due to competition	SUPPORTED	37203 has 78 shops (most in Nashville) but only 3.65 avg stars. 71.8% survival rate.	Avoid downtown. High foot traffic does not compensate for brutal competition and low quality perception.
H2: High-review shops regress toward 3.5 to 4.0 stars	DISPROVED	Shops with 500+ reviews score 4.12 avg stars, the highest of any review bucket.	Quality and popularity rise together. Build quality first. Reviews and ratings will follow.
H3: 7-day shops earn better ratings than restricted-schedule shops	DISPROVED	Mon-Sat shops score 4.26 vs 3.57 for 7-day shops. A difference of 0.69 stars.	Operate 6 focused days with exceptional quality. Do not trade quality for availability.

Section 10
SQL Techniques Learned and Applied

10. SQL Techniques Learned and Applied

This project provided practical application of a comprehensive range of SQL techniques across 18 notebook cells. The following table documents each technique, its purpose in this project, and the cell in which it was applied:

SQL technique	Purpose in this project	Cells applied
SELECT, FROM, WHERE	Basic query structure, row selection, table access	All cells
LIKE with LOWER()	Case-insensitive coffee category filter	5 to 18
COUNT(*), AVG(), SUM(), MIN(), MAX()	Aggregate statistics across datasets	1, 3, 5 to 11
ROUND()	Formatting decimal output to 2 to 3 significant figures	5 to 18
GROUP BY	Aggregating results by star rating, zip code, schedule type	4, 7 to 11, 13 to 18
ORDER BY ... DESC/ASC	Ranking results by rating, review count, score	6, 8, 10, 11, 14 to 18
HAVING	Filtering aggregated results (min shop count per zip)	8, 11, 14, 16 to 18
LIMIT	Restricting output to top N results for readability	6, 14 to 18
CASE WHEN ... THEN ... ELSE ... END	Bucketing continuous variables, conditional labelling	5, 9, 10, 11, 13, 14, 17
CAST(col AS STRING)	Converting bigint postal_code to correct string type	8, 11, 14, 16 to 18
UNION ALL	Combining multiple SELECT results into one output	1
SUM() OVER()	Window function for percentage calculation within GROUP BY	7
PERCENTILE(col, 0.5)	Computing median and quartile values	12, 14
STDDEV()	Standard deviation for spread analysis	12, 14
CORR()	Pearson correlation between two continuous variables	15
LOG10()	Logarithmic transformation for Quality Score metric	16
INNER JOIN ... ON	Cross-table analysis joining business and review tables	18
ABS()	Absolute value for sorting by magnitude of star gap	18
DESCRIBE TABLE	Column profiling and data type inspection	2

End of Report | CoffeeKing: SQL for Data Science Capstone Project | Savio Kenneth Myers | June 3, 2026